Euclid's Elements — Book I
Proposition 23

CGJteamLab

On a given straight line, and at a given point on it, to construct a rectilinear angle congruent to a given rectilinear angle.

1 Introduction

Euclid's Proposition I.23 asks for the transport of a given angle to a prescribed ray.

In Euclid's development this is a construction theorem. A suitable triangle is reconstructed from three side lengths using Proposition I.22, and Proposition I.8 (SSS) is then used to identify the required angle.

In the Hilbert reconstruction used by CGJteamLab, the logical status is different. Hilbert's congruence axiom III.4 already states that an angle can be constructed on a prescribed side of a prescribed ray, with uniqueness of the resulting ray. Consequently the formal Proposition I.23 is a short wrapper over the existing Hilbert infrastructure.

2 Euclid's Construction

Let a rectilinear angle be given, and let a point on a given straight line determine the ray on which the new angle is to be constructed.

Euclid chooses points on the two arms of the given angle, thereby forming a triangle. He then uses Proposition I.22 to construct a second triangle with the same three side lengths. Proposition I.8 (SSS) implies that the corresponding included angles are congruent.

Thus the Euclidean dependency chain is

$$\text{I.22} \longrightarrow \text{triangle with prescribed sides} \longrightarrow \text{I.8 (SSS)} \longrightarrow \text{I.23.}$$

Read literally, this route inherits the continuity issue hidden in Euclid's circle intersections. In the present project that issue has already been repaired in Proposition I.22 by an explicit application of circle-circle continuity.

3 Hilbert III.4

Hilbert makes angle transport primitive at the level of the congruence axioms.

Axiom III.4 says that, given a nondegenerate angle, a prescribed ray, and a choice of side of its supporting line, there exists exactly one ray on that side forming a congruent angle.

In the current Hilbert layer this is represented by

```
HilbertCongruence.angle_construction
```

Its conclusion supplies a point on the chosen side, the required angle congruence, and a uniqueness statement expressed through `HilbertSameRay`.

4 Formal Statement

The public theorem is deliberately weaker than the full Hilbert axiom because Euclid I.23 requires only existence, not uniqueness.

```
theorem euclid_proposition_23
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C D E S : Geo.Point)
    (hABC : ¬ PrimCollinear Geo A B C)
    (hDE : D ≠ E)
    (base : Geo.Line)
    (hDbase : HilbertIncidence.OnLine D base)
    (hEbase : HilbertIncidence.OnLine E base)
    (hSbase : ¬ HilbertIncidence.OnLine S base) :
    ∃ F : Geo.Point,
      HilbertSameSide Geo F S base ∧
      Geo.AngleCongruent A B C D E F
```

The source angle is $\angle ABC$. The ray DE is the prescribed initial ray of the new angle. The point S, chosen off the supporting line, specifies on which side the new ray is to lie.

The conclusion supplies a point F such that

$$\angle ABC \cong \angle DEF,$$

with F on the prescribed side of the base line.

5 Proof Architecture

The proof contains no new geometric construction.

It invokes `HilbertCongruence.angle_construction`, obtains a witness F, retains the same-side condition and the angle-congruence condition, and discards the uniqueness component because it is stronger than the Euclidean conclusion.

The complete formal dependency is therefore

$$\text{Hilbert III.4} \longrightarrow \texttt{angle\_construction} \longrightarrow \texttt{euclid\_proposition\_23}.$$

6 The Euclidean Route and the Hilbert Route

6.1 The Euclidean Route

Euclid reduces angle transport to triangle construction.

Proposition I.22 supplies a triangle with three prescribed sides. Proposition I.8 then identifies the corresponding angle by SSS. Thus angle transport is derived from segment construction, circle intersection, and triangle congruence.

6.2 The Hilbert Route

Hilbert reverses the foundational emphasis.

Angle transport is already available as an axiom of congruence. The theorem corresponding to Euclid I.23 therefore does not need Proposition I.22, SSS, circle construction, or continuity.

6.3 Logical Comparison

This is not merely a shorter proof of the same dependency chain. The dependency graph itself changes.

Euclid has

$$\mathrm{I}.22 + \mathrm{I}.8 \longrightarrow \mathrm{I}.23.$$

Hilbert has

$$\mathrm{III}.4 \longrightarrow \mathrm{I}.23.$$

The Book I sequence and the formal dependency order therefore diverge at this point.

7 Why Proposition I.22 Is Not Used Formally

The file for I.23 occurs after Proposition I.22 in the Book I sequence, but the Hilbert proof does not logically depend on I.22.

Reusing I.22 would deliberately reconstruct Euclid's historical proof on top of a system that already assumes the stronger angle-construction principle.

For the project this distinction is useful. The Book I files preserve the sequence of Euclid's propositions, while their formal dependency graph records what follows from the chosen Hilbert foundation. Those two orders need not coincide.

8 Uniqueness

Euclid's proposition asks only that a congruent angle be constructed.

Hilbert III.4 gives more: on the chosen side of the base line the resulting ray is unique.

The formal wrapper intentionally omits this extra conclusion. If another point F' lies on the same chosen side and determines the same angle, III.4 states that the ray from the new vertex through F' is the same as the ray through the constructed point.

This stronger result remains available directly from the Hilbert interface.

9 No New Book Zero Lemma

Unlike Proposition I.22, Proposition I.23 exposes no missing low-level geometric operation.

I.22 required a public witness-based predicate for the phrase "two segments together are greater than a third." I.23 requires no analogous addition to Book Zero because the exact construction is already present in the Hilbert congruence layer.

This is an important diagnostic outcome: not every Euclidean proposition should generate new helper infrastructure. When the chosen foundation already contains the proposition's mathematical content, the correct Book I implementation can be only an interface theorem.

10 Dependency Summary

The formal input consists of

- a nondegenerate source angle $\angle ABC$;
- a nondegenerate prescribed ray DE;
- a supporting line containing D and E;
- a point S selecting one side of that line.

Then

$$\begin{gathered}\text{nondegenerate angle } ABC\\ \text{prescribed ray } DE\\ \text{side selector } S\\ \downarrow\\ \texttt{HilbertCongruence.angle\_construction}\\ \downarrow\\ F \text{ on the prescribed side}\\ \angle ABC \cong \angle DEF\\ \downarrow\\ \texttt{euclid\_proposition\_23}.\end{gathered}$$

No continuity principle, segment-sum construction, SSS theorem, or Proposition I.22 is needed in the formal proof.

11 Formalization Notes

The point S is not part of Euclid's informal statement. It is the formal device selecting one of the two sides of the supporting line. Without this datum there are two mirror-image solutions.

The hypotheses that D and E lie on the base line and that $D \neq E$ specify the prescribed ray.

The hypothesis that A, B, C are not collinear guarantees that the source angle is nondegenerate.

The wrapper returns a point F rather than an abstract ray because rays in the project are represented through points and the relation `HilbertSameRay`. This matches the point-based style of the Hilbert interface.

12 Conclusion

Proposition I.23 is a useful contrast with Proposition I.22.

I.22 required substantial synthetic segment arithmetic and an explicit continuity theorem. I.23, although Euclid proves it by invoking I.22 and SSS, is immediate in Hilbert geometry.

The resulting Lean theorem is therefore intentionally small. Its brevity is not a missing proof: it records the fact that Hilbert III.4 already contains the existence statement of Euclid I.23, together with a stronger uniqueness assertion.

This is exactly the kind of distinction the reconstruction is intended to expose: the propositions of Book I remain the visible mathematical milestones, while the Hilbert layer reveals which of those milestones are derived results and which have effectively been moved into the foundational interface.